

Project Initialization and Planning Phase

Date	11 July 2026
TeamID	XXXXXX
Project Title	EduGenie-Google Gemini Powered Learning Assistant
Maximum Marks	5Marks

Project Proposal(Proposed Solution)Report

The proposed solution introduces EduGenie – Google Gemini Powered Learning Assistant, an AI-powered educational platform that enhances learning through intelligent academic assistance. The system uses Google Gemini 1.5 Pro and LaMini-Flan-T5 models to provide concept explanations, answer academic questions, generate quizzes, summarize educational content, and recommend personalized learning paths. The application delivers accurate, interactive, and personalized learning support through an easy-to-use web interface.

Project Overview

Topic	Description
Objective	Develop an intelligent AI-powered educational assistant that helps students understand concepts, answer academic questions, generate quizzes, summarize learning materials, and provide personalized learning recommendations.
Scope	The system supports concept explanation, question answering, quiz generation, text summarization, personalized learning paths, AI-based educational assistance, and a responsive web interface for students and educators.

Problem Statement

Topic	Description
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Description	Students often struggle to understand complex concepts, search across multiple resources, create study materials, and receive personalized guidance for effective learning. Existing platforms provide fragmented educational services instead of an Integrated AI learning solution.
Impact	An AI-powered educational assistant improves learning efficiency, reduces study time, enhances conceptual understanding, and provides personalized academic support for students.

ProposedSolution

Topic	Details
Approach	Use Google Gemini1.5Pro,LaMini-Flan-T5,Fast API,and modern web technologies to process user requests and generate AI-powered educational content, including explanations, answers, quizzes, summaries, and personalized learning recommendations.
Key Features	• Concept Explanation • Question Answering(QnA) • Quiz Generation • Text Summarization • Personalized Learning Path Recommendations • AI-powered Educational Assistance • Responsive Web Interface

ResourceRequirements

ResourceType	Description	Specification/Allocation
Hardware		
Processor	CPU	Intel Core i5/Ryzen 5 or higher
Memory	RAM	8GB or above

Storage	Disk Space	10 GB free SSD space
Software		
Framework	Application Framework	Fast API
Libraries	Python Libraries	Google Generative AI SDK,Transformers, Pydantic, Jinja2, Uvicorn
Development Environment	IDE	Visual Studio Code
Data		
Dataset	Reference Concepts & Audio Samples	Learning topics, user queries, quizzes, summaries, and personalized learning recommendations